

PURBANCHAL UNIVERSITY

2024

Master of Computer Application (M.C.A.)/First Semester/ *Final*

Time: 03:00 hrs.

Full Marks: 80 / Pass Marks: 32

MCA102: Software Engineering (*New Course*)

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

1. What is software project planning and why it is required for the development of software? Explain COCOMO model in detail. 4+8
2. Explain the importance of software testing. Differentiate between White Box and Black Box testing in detail.
3. How can we achieve high reliability in software? Explain the parameters of software evaluation. 4+8

Group B

Answer SEVEN questions.

7×8=56

4. What is software reengineering? What type of ethical values should software engineer be possessed? 2+6
5. What is software configuration management? Explain different standards of SCM in detail. 2+6
6. Differentiate between OOA and OOD. Explain MVC architecture with suitable example. 2+6
7. How quality assurance is different from quality control? How can we maintain the CMMI Level for the software process improvement? 3+5
8. Why SRS is required in software engineering. Explain functional and non-functional requirements. 2+6
9. What do you mean by reused based software engineering? What are the characteristics of maintainable software? 4+4
10. List out various types of emerging technologies used in software development. How is Big Data used in software? 2+6

Contd. ...

(2)

11. Write short notes on any TWO:

2×3=6

- (a) Agile Manifesto
- (b) Cyclomatic complexity
- (c) Version control

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